

WEEK-7 TASK

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Project Title: Customer Churn Prediction using Decision Tree

Week: Week 7

Task Name

Customer Churn Prediction using Decision Tree

Abstract

Customer churn prediction is a machine learning technique used to identify customers who are likely to stop using a service. A Decision Tree analyzes customer information such as usage, subscription period, and payment details to classify customers as likely to churn or remain active.

Introduction

Customer retention is important for businesses because losing customers can affect revenue and growth. A Decision Tree provides an easy-to-understand approach for predicting churn by dividing customer data based on important features.

Objective:

Build a Decision Tree model to predict whether a customer is likely to leave a service. The system should analyze customer attributes and classify them into churn and non-churn categories.

Main Steps

1. Prepare Customer Data

- Customer age
- Monthly usage
- Subscription duration
- Monthly charges
- Payment method
- Customer support requests

2. Build a Decision Tree

- Select the important feature.
- Calculate the best splitting condition.
- Divide the customer data into groups.
- Repeat the process recursively.
- Stop when the required tree depth is reached.

3. Train the Prediction Model

- Divide data into training and testing sets.
- Train the Decision Tree using training data.
- Apply the learned rules to new customers.
- Generate churn predictions.

4. Evaluate Prediction

The basic accuracy formula is:

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

Where:

- Correct Predictions = Number of correctly classified customers
- Total Predictions = Total number of customers tested

Interpretation:

- High accuracy → model predicts customers effectively
- Medium accuracy → model requires further improvement
- Low accuracy → model needs better features or training data

Expected Workflow

Customer Data

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Data Preprocessing

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Feature Selection

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Training and Testing Split

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Build Decision Tree

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Train Prediction Model

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Predict Customer Churn

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Evaluate Accuracy



Churn / Non-Churn

Conclusion

Decision Tree provides a simple and interpretable method for predicting customer churn. By analyzing customer characteristics and applying recursive data splitting, the model can identify customers who may leave a service. Building the model helps understand classification, feature selection, prediction, and model evaluation.